

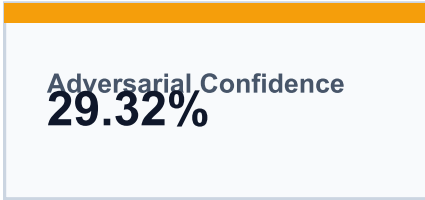
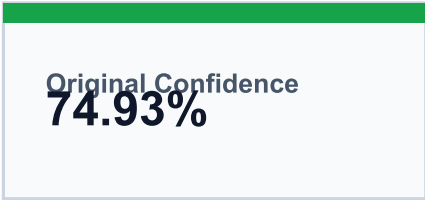
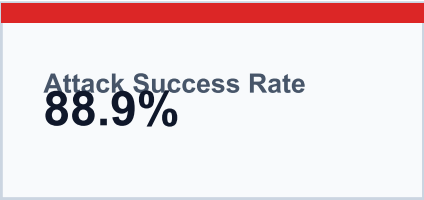
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: microsoft/resnet-18

Attack Type: FGSM

Images Evaluated: 9



Threat Level Assessment

HIGH RISK

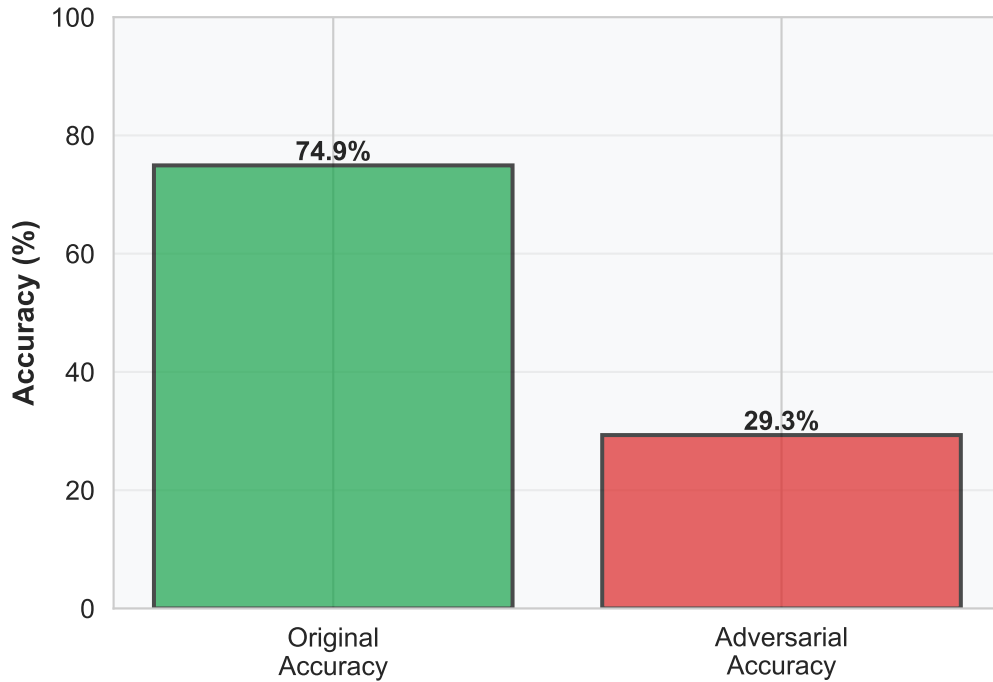
The FGSM attack altered model predictions on 88.9% of the evaluated images. Average confidence shifted from 74.93% to 29.32%, indicating the current robustness posture is high risk.

Assessment Narrative

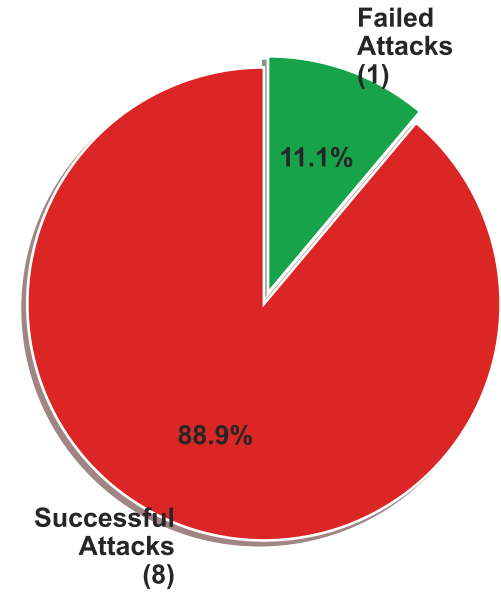
Successfully executed FGSM attack on model microsoft/resnet-18 using 9 test images. Attack success rate: 88.9%. Average original accuracy: 74.93%, Average adversarial accuracy: 29.32%. The attack successfully fooled the model in 8 out of 9 cases.

Detailed Analysis

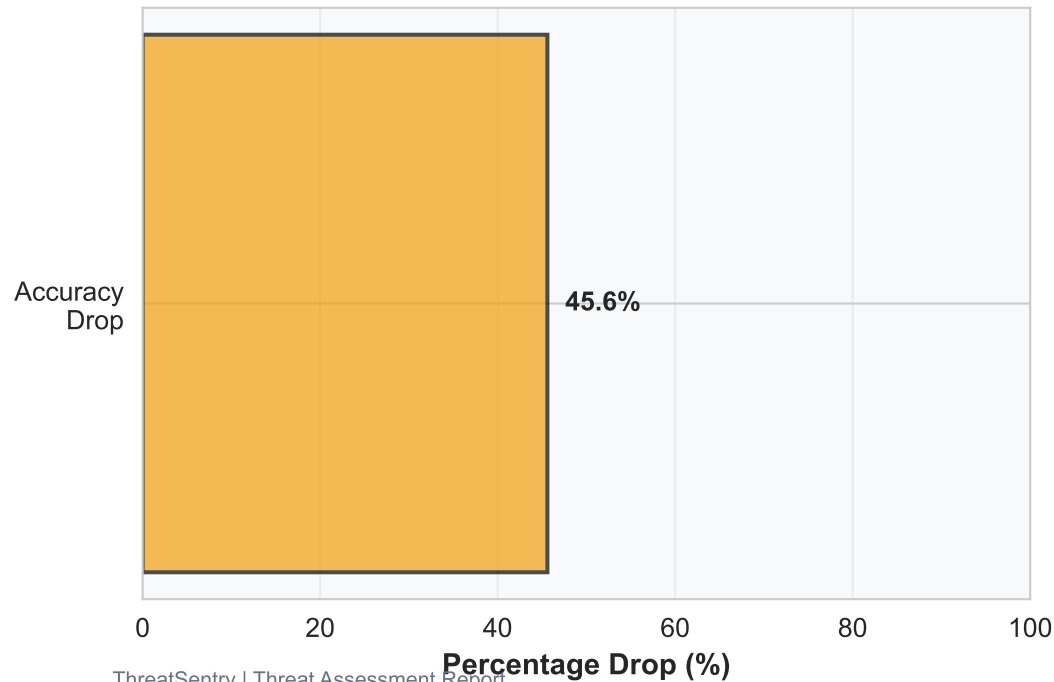
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

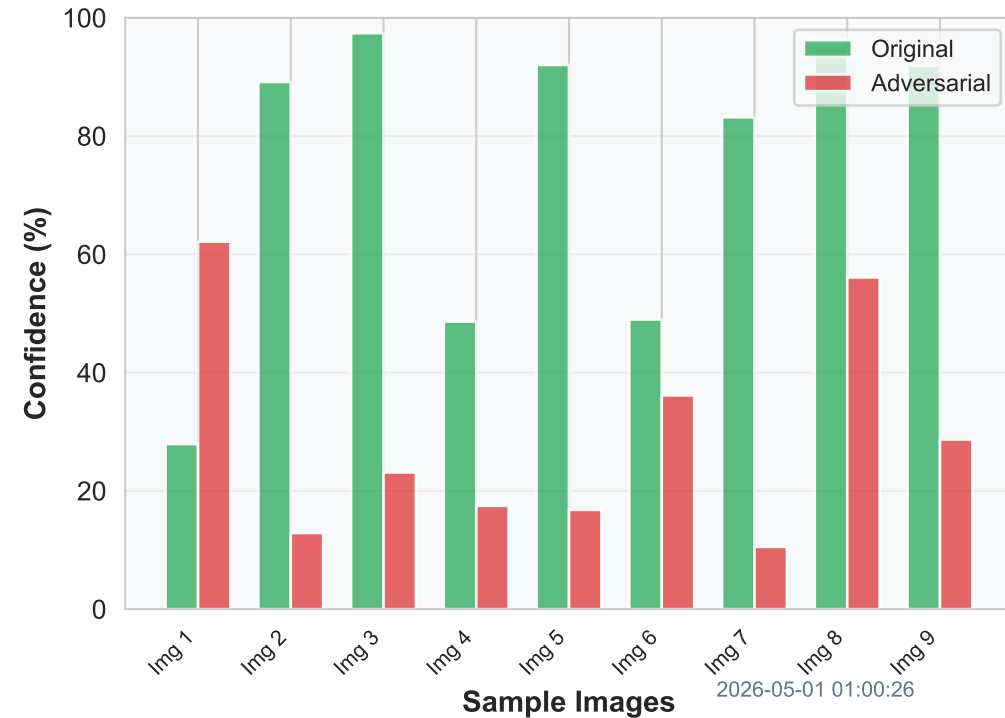


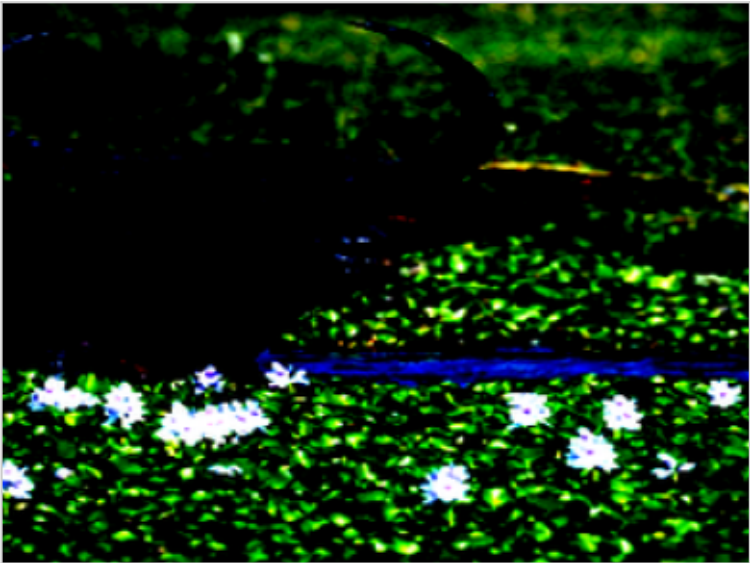
Image Evidence - Original vs Adversarial

Images 1-2 of 9

Original: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis
(97.4%)



Adversarial: American alligator, Alligator mississippiensis
(23.1%)
ATTACK SUCCESSFUL



Original: titi, titi monkey
(89.2%)



Adversarial: platypus, duckbill, duckbilled platypus, duck-billed platypus, Ornithorhynchus anatinus
(12.9%)
ATTACK SUCCESSFUL

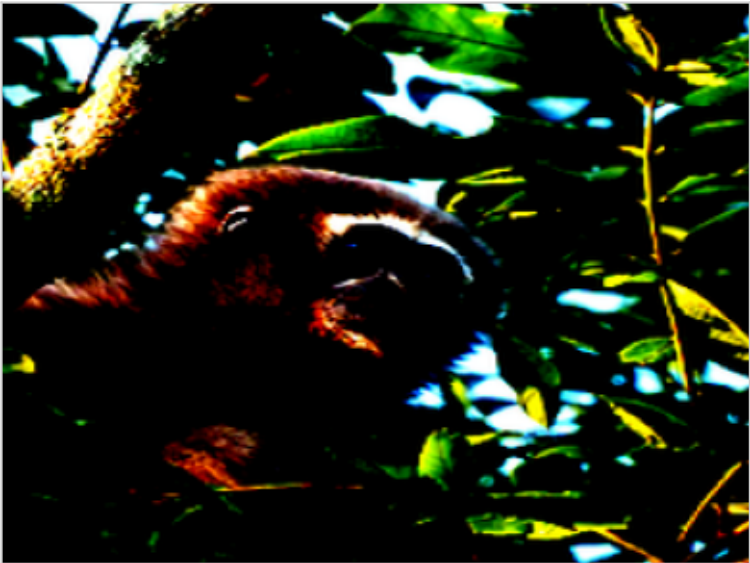


Image Evidence - Original vs Adversarial

Images 3-4 of 9

Original: ibex, Capra ibex
(83.2%)



Adversarial: water snake
(10.5%)
ATTACK SUCCESSFUL



Original: coyote, prairie wolf, brush wolf, Canis latrans
(91.9%)



Adversarial: warthog
(28.7%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 5-6 of 9

Original: tiger, *Panthera tigris*
(95.1%)



Adversarial: tiger, *Panthera tigris*
(56.1%)
ATTACK BLOCKED



Original: dhole, *Cuon alpinus*
(92.0%)



Adversarial: American alligator, *Alligator mississippiensis*
(16.8%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 7-8 of 9

Original: Indian elephant, *Elephas maximus*
(49.0%)



Adversarial: Arabian camel, dromedary, *Camelus dromedarius*
(36.2%)
ATTACK SUCCESSFUL



Original: indri, indris, *Indri indri*, *Indri brevicaudatus*
(48.6%)



Adversarial: spider web, spider's web
(17.5%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 9-9 of 9

Original: hog, pig, grunter, squealer, *Sus scrofa*
(27.9%)



Adversarial: water buffalo, water ox, Asiatic buffalo, *Bubalus bubalis*
(62.1%)
ATTACK SUCCESSFUL



Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model microsoft/resnet-18 using 9 test images. Attack success rate: 88.9%. Average original accuracy: 74.93%, Average adversarial accuracy: 29.32%. The attack successfully fooled the model in 8 out of 9 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques